

Product Definition, Scope, Roles & Principles

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Document role. This is document 00 of the HelixVPN master technical specification set under docs/research/mvp/final/. It establishes the *invariants every later document inherits*: what the product is and is not, who uses it, the three-role topology, the licensing/hosting stance, the per-phase scope boundary, the seven non-negotiable principles, the Mullvad-parity acceptance matrix, and — crucially — the **eight open architectural decisions (D1–D8)** (D1–D7 technical + D8 licensing, §7.2) that downstream documents (01-transport, 02-control-plane, 03-client-core, ...) must resolve. Where a decision is still open this document gives **options plus a recommendation** (§11.4.6 no-guessing, §11.4.66 decision discipline); it never silently picks a camp.

Evidence base. Cross-document synthesis of the 16 source research docs in docs/research/mvp/ (eleven independent LLM analyses 00,01_DSK,02_QWN, 03_ZAI,05_YB0,06_GRK,07_GMI,08_CPL,09_GCT,10_KMI,11_MST plus the five refined 04_VPN_CLD docs). Citations are inline by id, e.g. [04_ARCH §1] = 04_VPN_CLD/HelixVPN-Architecture-Refined.md §1, [05_YB0] = the operator mandate brief, [research-masque] = the MASQUE research note.

Status of this document: SPEC-ONLY. It describes *what to build and why*; it does not build it. Two-to-three refinement passes follow.

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1. Executive product definition

HelixVPN is a self-hostable overlay network with a privacy-VPN front end.

The one-sentence pitch [04_ARCH §1]:

Cloudflare Tunnel + WARP, rebuilt as Tailscale-style coordination, with Mullvad's obfuscation stack, fully self-hostable, on one shared codebase.

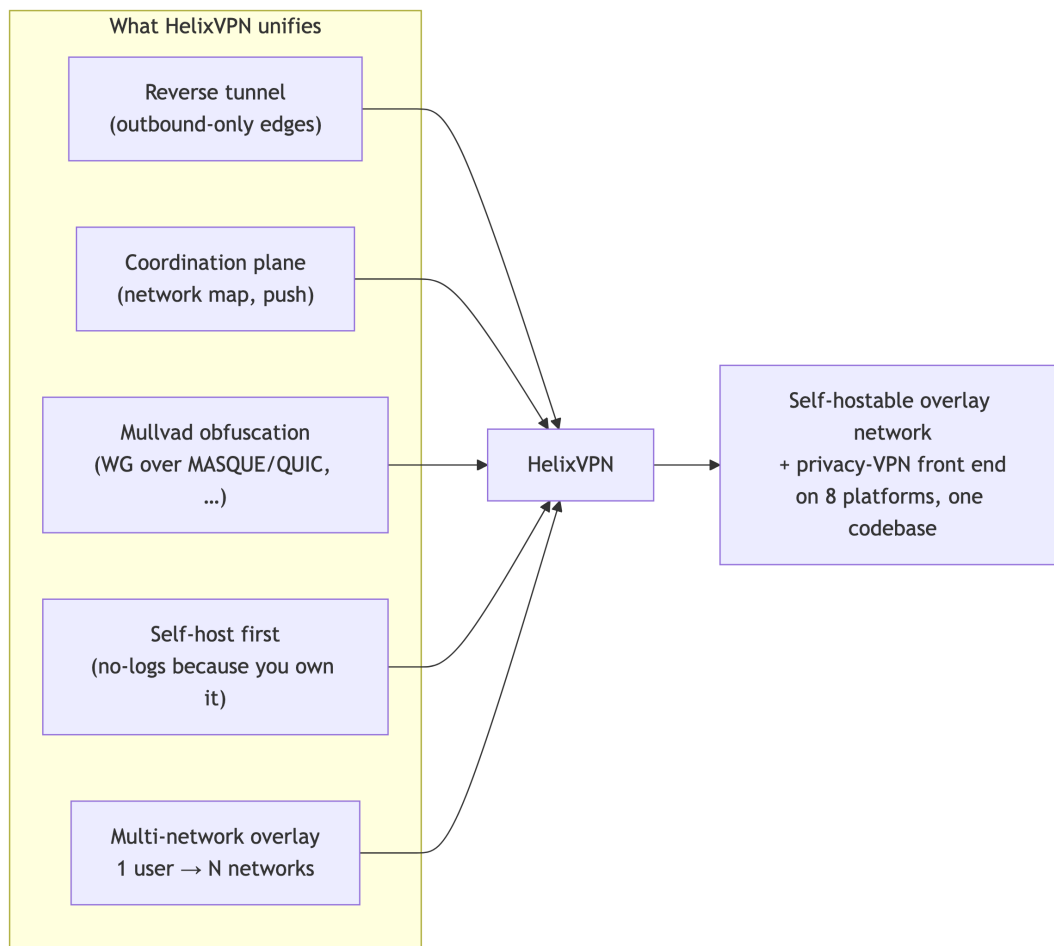
Decomposed, that pitch names four lineages, each of which HelixVPN deliberately unifies:

Lineage borrowed from	What HelixVPN takes	Source
Cloudflare Tunnel + WARP	the <i>connector dials out</i> model (no inbound port-forward) and MASQUE/QUIC transport	[00], [04_ARCH §0/ §3.3]
Tailscale / Headscale	the <i>coordination & desired-state</i> "network map" model (push-don't-poll, ACL-compiled topology)	[04_ARCH §4.4], plurality of analyses
Mullvad	the <i>obfuscation + privacy bar</i> (QUIC/MASQUE, Shadowsocks, UoT, LWO, DAITA, multi-hop, kill-switch, no-logging, anonymous accounts)	[04_ARCH §6], [02_QWN], [11_MST]

Lineage borrowed from	What HelixVPN takes	Source
NetBird	the <i>closest OSS shape</i> (WireGuard + management/signal server, self-hosted) as an architectural sanity check	[04_ARCH §1.3]

The **founding constraint** that gave rise to the project [05_YB0], [00]: a remote user must obtain full, policy-scoped access to one *or many* internal / home / lab networks **without any inbound port-forward on those networks**. Internal hosts dial *outbound* to a public gateway (a reverse tunnel); the gateway relays and routes. The headline differentiator over every incumbent is **1 user → N joined private networks** — a multi-network, bidirectional, policy-routed gateway — which no self-hostable product ships today [04_ARCH §1.3].

Reconciled (§11.4.35, 2026-06-26) — UNVERIFIED (§11.4.99). "no self-hostable product ships [a 1→N multi-network policy-routed gateway] today" is a **competitive-market claim**, not an architectural fact of HelixVPN. It is plausible from the prior-art map (§6) but **MUST** be re-verified against current product feature sets in the §11.4.99 latest-source pass before any external-facing use. This matches the marker already carried by v01-product/product-vision-and-positioning.md §2.2/§8 (R-V4). The *architectural* fact — that HelixVPN's design delivers 1→N policy-routed networks — is cited and is **not** UNVERIFIED.



2. What HelixVPN is — and is explicitly NOT

Pinning the negative space is as load-bearing as the positive definition: it keeps every downstream document from drifting into adjacent products.

2.1 HelixVPN IS

1. A **WireGuard-cryptographic-core overlay** where obfuscation/transport is a *pluggable layer underneath* WireGuard, never a crypto fork [04_ARCH §2/§3].
2. A **three-role system** — Connector, Gateway, Client — with a strict control-plane / data-plane split [04_ARCH §1.1/§2].
3. **Self-hostable by one person** (single rootless-Podman pod), with the *same images* scaling to an HA multi-region fleet [04_ARCH §2 principle 6, §10].
4. A **cross-platform app suite** on a shared codebase, targeting iOS, Android, **Aurora OS**, **HarmonyOS NEXT**, Windows, Linux, macOS, and Web [05_YB0].

5. An **event-driven, push-based** control plane: state changes propagate as events; agents reconcile to a declared desired-state [04_ARCH §4.3/ §4.4].
6. A **no-logging-by-construction** privacy product: only aggregate counters and ephemeral presence persist; no durable connection / traffic / packet table ever exists — CI-enforced [04_ARCH §4.5/§7].

2.2 HelixVPN is NOT

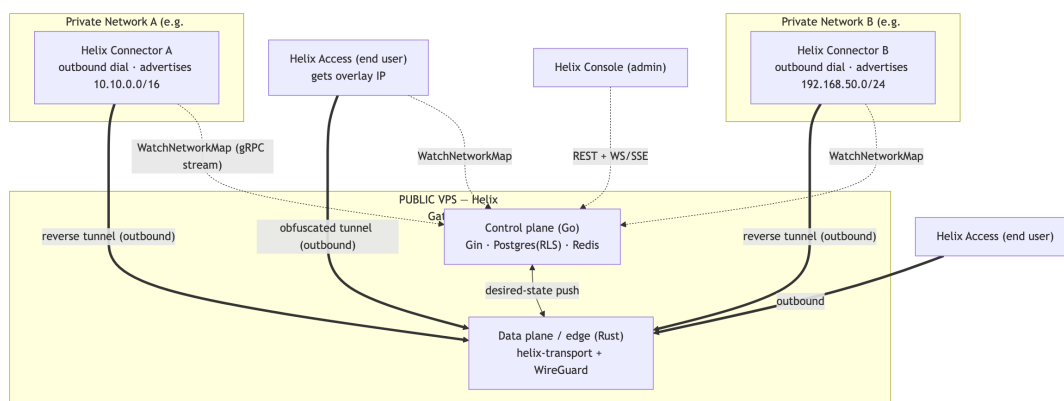
NOT	Why it is excluded	Where the boundary is enforced
NOT a forked or re-rolled WireGuard crypto.	The crypto core is the audited WireGuard Noise IK construction. We change only how the <i>encrypted datagrams look on the wire</i> .	Transport spec 01; CI lint forbids touching WG crypto primitives.
NOT a separate "QUIC protocol" alongside WG.	Mullvad's QUIC mode <i>is</i> WireGuard tunneled over MASQUE/HTTP-3, not a different protocol [04_ARCH §0], [research-masque].	D1 below; transport spec 01.
NOT a full system VPN in a browser.	Browsers cannot open a TUN device. The Web build is Console (management) + an <i>optional</i> in-page WASM MASQUE proxy that proxies the browser's own traffic only [04_ARCH §5.7].	Client/app spec 03/05; §10 scope below.
NOT a logging / lawful-intercept / DLP appliance.	No-logging is an architectural build property, not a toggle. <i>Control</i> actions are audited; <i>traffic</i> never is.	Data-model spec 02; CI schema-lint.
NOT a generic SDN / service-mesh / L7 API gateway.	HelixVPN operates at L3 (IP overlay) with L4 port policy. It is not Envoy/Istio and does not terminate application TLS for inspection.	Policy spec; §2.1 scope.
NOT a consumer "free VPN" ad-funded service.	The default deployment is self-hosted; a managed offering is an optional later SKU, not the product's reason to exist.	§7 below.
NOT dependent on inbound port-forwarding on	The reverse-tunnel insight from the original brief is foundational.	Principle 3 (§8); Connector spec.

NOT	Why it is excluded	Where the boundary is enforced
any joined network.		

§11.4.6 honesty marker. Two product claims have *hard physical limits* and **MUST** be stated to users without hand-waving [04_ARCH §5.7]: (a) **"fully responsive web app"** ≠ **system VPN** — it is the Console + a browser-scoped proxy; (b) **iOS Network Extension ~15 MB working-set ceiling** is a measured historical constraint that shapes the core-language decision (D2) and is verified on-device in Phase 0 gate G3, never assumed.

3. The three roles: Connector ⇔ Gateway ⇔ Client

The entire system is three roles and three control relationships [04_ARCH §1.1], [05_YB0], consensus across all 10 analyses.



3.1 Connector (network-side agent)

- A **first-party agent installed on a host inside each private network**.
- **Dials outbound** to the Gateway — *no inbound port-forward, ever* [05_YB0].
- Authenticates (device cert + enrollment token), then **advertises the CIDRs** that network exposes (route.advertised).
- Routes Gateway-bound traffic *into* its LAN and LAN responses *back out*.
- Runs **headless** (daemon) with an optional slim config UI; on appliance hardware it can run on Android/embedded [04_ARCH §1.2/§5.4].
- Shares the **same Rust core** as the Client, in *advertise/route* mode rather than *capture-this-device's-traffic* mode [04_ARCH §5.4].

3.2 Gateway (public VPS — the rendezvous hub)

- The only role with a public address. Two strictly separated planes:
 - **Control plane (Go/Gin/Postgres/Redis)** — source of truth for identity, devices, networks, routes, policy; distributes desired-state in real time; **never sits in the packet path** (principle 1).
 - **Data plane / edge (Rust + kernel WireGuard)** — terminates obfuscated transports, unwraps to WireGuard, applies the per-peer verdict map, routes between users and the networks they are authorized to reach.
- Authenticates connectors and clients, holds the routing/policy table, relays/routes traffic [04_ARCH §1.1/§4].

3.3 Client (end user — Mullvad-style)

- The **Access app**. Dials into the Gateway, receives an **overlay IP**, and reaches the *subset of joined networks its policy allows*.
- Can also use the Gateway as a **plain privacy exit** (full-tunnel to the internet) — the Mullvad use case [04_ARCH §1.1].
- Drives the *same Rust core* as the Connector (capture mode) + a per-platform tunnel shim.

3.4 The control relationships (one sentence each)

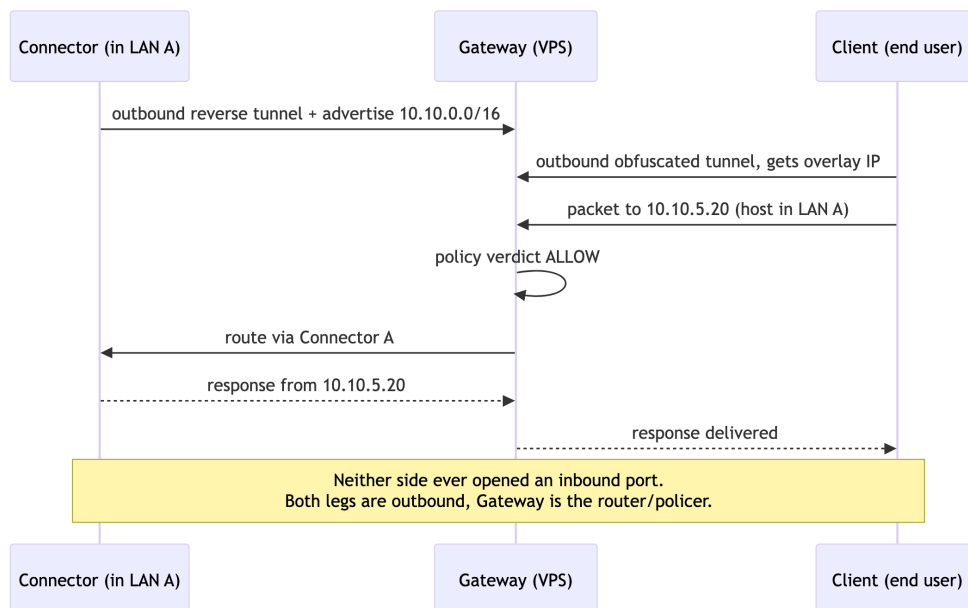
Relationship	Channel	Direction	Purpose
Connector → Gateway control	gRPC WatchNetworkMap (server-stream over HTTP/2 or HTTP/3)	Connector dials out	receive snapshot+deltas; advertise prefixes
Client → Gateway control	gRPC WatchNetworkMap	Client dials out	receive <i>policy- filtered</i> peer/route map
Console → Gateway control	REST (Gin) + WS/SSE	admin dials out	CRUD + live updates
Connector/ Client → Gateway data	obfuscated transport → WireGuard	edge dials out	the encrypted packet path (fail- static)

4. "Two-way" and "multi-network" — precise meaning

The operator brief uses two phrases that the original Home_VPN.md did not fully deliver [04_ARCH §0], [05_YB0]. Their *exact* engineering meaning:

4.1 "Two-way VPN"

Both halves dial outbound and the Gateway routes between them. It does **not mean a single point-to-point tunnel:**



The "two ways" are the **network-side leg** (Connector → Gateway) and the **user-side leg** (Client → Gateway). The Gateway *stitches and polices* between them. This is the core correction over the original one-way "reverse tunnel for my home" model [04_ARCH §0/§1.1].

4.2 "Multi-network"

1 user → N joined private networks, each policy-scoped. Multiple Connectors each advertise their own CIDRs; the Gateway compiles them into one *policed overlay*; a single user reaches the subset of those networks its ACL grants. This implies three first-class problems the spec set **MUST** solve [04_ARCH §3.4]:

1. **Overlapping RFC1918 ranges.** Two connectors can both expose 192.168.1.0/24. The overlay must present each as a *distinct* address space so they never collide — this is **D4** (§11), a v1 must-decide.

2. **Per-user ACL routing.** Default-deny; group:contractors → net:warehouse:554/tcp compiles to per-peer AllowedIPs + an nftables/eBPF verdict map.
3. **Split horizon.** Connectors cannot reach each other, and clients cannot reach networks they are not granted, unless policy says so. Microsegmentation is the default [04_ARCH §3.4].

5. Personas and the three app classes

One design system, one Dart UI core, one generated API client across all three apps. Access + Connector additionally share one Rust VPN/transport core [04_ARCH §1.2/§5].

App class	Persona	Primary jobs	Platforms	Cores used
Helix Access	End user ("I want into my lab / a privacy exit")	connect, pick exit/network, toggle obfuscation, kill-switch, split-tunnel	iOS, Android, Aurora, HarmonyOS, Windows, Linux, macOS (Web = limited)	helix-ui + helix-core + tunnel shim
Helix Connector	Network operator ("expose my LAN safely")	onboard a network, advertise CIDRs, set local ACLs, run headless	Linux/ Windows/ macOS daemon + optional UI; Android/ embedded for appliances	helix-core (advertise/route mode) + optional helix-ui
Helix Console	Admin ("manage tenants/users/devices/policy")	tenants, users, devices, networks, routes, policies, audit, billing-optional	Web (responsive) + Desktop (same Flutter build)	helix-ui (admin flavor) + API client only — no core, no tunnel

5.1 Persona detail

- **End user (Access).** Wants the Mullvad experience: one big connect button, automatic obfuscation, kill-switch, no account email required (anonymous device token). Memory- and battery-sensitive (mobile) [04_ARCH §1.2/§5.6].

- **Network operator (Connector).** Wants "install one agent inside the LAN, it dials out, done." Cares about which prefixes are advertised, local ACLs, and *not* opening a router port [05_YB0], [04_ARCH §1.1].
- **Admin (Console).** Wants tenant/user/device/policy CRUD, a live topology view, an audit trail of *control* actions, and (optionally) multi-tenant billing. Runs in a browser or as a desktop app — no tunnel involved [04_ARCH §1.2/§5.4].

5.2 The single-tree flavoring contract

All three apps build from one Flutter tree via a flavor endpoint `runHelixApp(flavor, home, capabilities)` [04_UI], where `flavor ∈ {Access, Connector, Console}`; Console is the *only* build that omits `core_ffi` (no Rust tunnel core). This contract is fixed here and detailed in client doc 03/05.

6. Prior art and positioning

System	Self-host	Obfuscation	Multi-network overlay	1st-party cross-platform apps	What HelixVPN borrows
Mullvad	× (service)	✅ best-in-class (QUIC/MASQUE, Shadowsocks, UoT, LWO, DAITA)	× (privacy exit only)	✅	the obfuscation + privacy bar
Tailscale	partial (Headscale)	×	✅ (mesh + subnet routers)	✅	the coordination / network-map model
NetBird	✅	limited	✅	✅	closest OSS shape (WG + mgmt/signal)
Cloudflare Tunnel + WARP	×	MASQUE	✅	✅	connector-dials-out + MASQUE
Twingate / Zscaler	×	n/a	✅ ZTNA	✅	the ZTNA policy model
HelixVPN	✅		✅	✅ on 8 platforms,	the self-hosted <i>union</i>

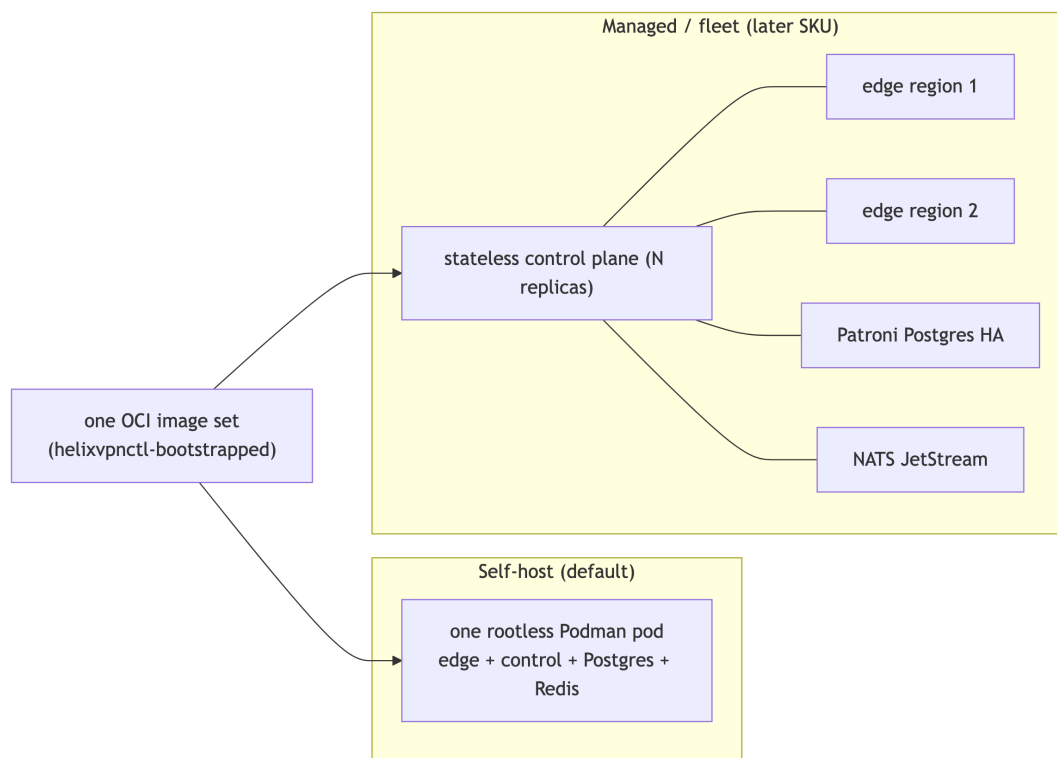
System	Self-host	Obfuscation	Multi-network overlay	1st-party cross-platform apps	What HelixVPN borrows
		✅ full Mullvad-parity stack		one codebase	of all of the above

Differentiators to lead with [04_ARCH §1.3]: (1) self-hosted *and* Mullvad-grade obfuscation including MASQUE/QUIC; (2) genuine 8-platform reach including Aurora + HarmonyOS that no incumbent ships; (3) one shared Rust+Flutter codebase; (4) event-driven real-time control plane.

7. Hosting model, licensing & commercial stance

7.1 Hosting model — settled

Self-hosted / home-lab first; the same code can serve managed later [04_ARCH §2 principle 6, §13], consensus. Rationale: a privacy product's no-logs claim is *credible because you own the box*, not because a vendor promises it [SYNTHESIS §2]. Two deployment shapes, one image set:



7.2 Licensing (D8) — OPEN DECISION, recommendation given

Reconciled (§11.4.35, 2026-06-26): the licensing decision is **D8** in the decision register (§11). SPECIFICATION.md §9 and the decision-register already number it D8, and v01-product/product-vision-and-positioning.md §3.1 cites "[00 §7.2]" as D8's source — this section is therefore the **canonical definition** of D8 and it is summarised in §11.1.

The licensing model is *not yet settled* [04_ARCH §13] ("decide the licensing ... before public release"). Because it materially shapes the repository split (§11.4.28/.74 decoupled reusable components → own vasic-digital repos), it is surfaced here as decision **D8**, not a default.

Option	Description	Pros	Cons
L-A: AGPLv3 (copyleft)	strong network-copyleft; managed competitors must open their changes	maximal community-protection; aligns with WireGuard/Mullvad ethos	scares some enterprise self-hosters; complicates a future closed managed SKU
L-B: Source-available (e.g. BSL → Apache after N years)	source visible, commercial-use clause, time-delayed OSS	protects a future managed business; still auditable (privacy claim)	not OSI-"open"; community friction
L-C: Apache-2.0 / MIT (permissive) core + commercial Console/managed add-ons	open core, paid edges	broadest adoption; clean for the reusable Rust/Flutter cores reused by other Helix projects	a competitor can run a managed clone with no give-back

Recommendation: L-C (open-permissive reusable cores + L-B source-available for the managed/multi-tenant Console layer). Rationale: the three reuse pillars — helix-core (Rust), helix-ui (Flutter), helix-proto — are *designed to be reused by other Helix-ecosystem projects* [04_ARCH §11], [SYNTHESIS §6] and therefore want a permissive license; the *commercial surface* (multi-tenant billing, fleet orchestration) is exactly the BSL-protectable layer. This split also satisfies §11.4.28 (owned submodules are decoupled, reusable, project-not-aware) — a permissive core is the most

reusable. **This decision is OWNER-GATED** (§11.4.66) before public release; until resolved, all repos carry an explicit `LICENSE.PENDING` marker, never an assumed license (§11.4.6).

7.3 No-logging as a hosting invariant

Independent of hosting shape, **no durable connection/traffic/packet table may exist** (§8 principle 7). For the managed SKU this is the *contractual* no-logs promise; for self-host it is *self-evident*. Either way it is CI-enforced (schema-lint), not a runtime toggle [04_ARCH §4.5/§7].

8. The seven non-negotiable principles

Verbatim from [04_ARCH §2], each with its engineering rationale and the downstream document that enforces it. These are **invariants**: a later document may add constraints but may not weaken any of these (mirrors the constitution inheritance rule §11.4.35).

P1 — Control plane and data plane are strictly separated

The Go control services **never sit in the packet path**. If the control plane is down, existing tunnels keep forwarding (**fail-static**). *Rationale*: a coordination outage must never become a connectivity outage; it also keeps the GC'd Go runtime off the hot path. *Enforced by*: control-plane spec 02 (Go emits desired-state only); edge spec (Rust forwards from a cached verdict map). *Anti-pattern forbidden*: any synchronous Go call on a per-packet basis.

P2 — WireGuard is the cryptographic core; transports are pluggable

Obfuscation is a swappable layer **under** WireGuard, never a fork of the crypto [04_ARCH §2/§3]. *Rationale*: the audited Noise IK / Curve25519 / ChaCha20-Poly1305 construction is the trust anchor; censorship resistance is a *wire-shape* problem solved beneath it (this is literally Mullvad's model, and why "QUIC" = WG-over-MASQUE, not a new protocol) [research-masque]. *Enforced by*: transport spec 01 (the Transport trait — see D1) + CI lint forbidding edits to WG crypto.

P3 — Outbound-only from edges

Connectors and Clients **always dial the Gateway**. No private network ever needs an inbound hole [05_YB0]. *Rationale*: the founding constraint; it is what makes "expose my LAN without touching my router" true and removes the largest attack surface (a public inbound port). *Enforced by*: Connector/Client spec; the Gateway is the only role that listens publicly.

P4 — Push, don't poll

State changes propagate over persistent channels as **events**; agents reconcile to a declared desired-state ("network map"), Tailscale-MapResponse-style [04_ARCH §4.4]. *Rationale*: polling/cron-restart loops (the original doc's model) cannot meet the **p99 < 1 s** convergence target and waste battery/CPU. *Enforced by*: control-plane spec 02 (WatchNetworkMap server-stream + Redis Streams event bus — see D3) + reconciler in helix-core.

P5 — One core per concern, reused everywhere

Rust transport/VPN core shared by Client, Connector, *and* Gateway edge; Go domain libraries shared across control services; Dart UI core shared across all three apps [04_ARCH §2/§5.5]. *Rationale*: the obfuscation code that *wraps* on the client is the *same code* that *unwraps* on the edge — one implementation, three consumers — eliminating drift and triplicated bugs. *Enforced by*: repo layout (§13); three reuse pillars (Rust core / Flutter UI / schema-generated clients).

P6 — Self-hostable by one person, scalable to many gateways

A single rootless-podman deploy for a homelab; the *same images* scale to an HA, multi-region fleet [04_ARCH §2/§10]. *Rationale*: the homelab user and the managed operator must not need different codebases; growth is a deployment topology change, not a rewrite. *Enforced by*: deploy spec (Podman quadlets, §11.4.76 containers submodule, §11.4.161 rootless).

P7 — No-logging by construction

The data plane keeps only **counters and ephemeral routing state**; no connection/content logs. Privacy is a *build property*, not a config toggle [04_ARCH §2/§7]. *Rationale*: a no-logs claim you can *grep the schema* for is stronger than a policy promise. *Enforced by*: data-model spec 02 (no connections/traffic/ packet durable table; live presence lives in TTL'd

Redis) + **CI schema-lint fails the build** if such a table appears [04_ARCH §4.5]. This composes with the constitution anti-bluff covenant (§11.4 / §11.4.69 captured-evidence).

Representative CI guard (illustrative skeleton, not the product)
— the mechanical teeth of P7:

```
// tools/schemalint/nolog.go – fails the build on a durable
// traffic table.
var forbiddenDurableTables = []string{"connections",
    "traffic", "packets", "sessions_log", "flows"}

func LintNoDurableTraffic(schema *pg.Schema) []Violation {
    var v []Violation
    for _, t := range schema.Tables {
        for _, banned := range forbiddenDurableTables {
            if strings.EqualFold(t.Name, banned) &&
                t.IsDurable() {
                v = append(v, Violation{Table: t.Name,
                    Rule: "P7-no-logging", Msg: "durable
                    traffic/connection table forbidden"})
            }
        }
    }
    return v
}
```

9. Mullvad feature-parity matrix (acceptance checklist)

This table is the **acceptance checklist for "all Mullvad power features"** [04_ARCH §6], [05_YB0]. Every row maps to a concrete component in a later document; nothing is aspirational. The "Phase" column ties each feature to the scope table in §10.

#	Mullvad feature	What it does	HelixVPN implementation	Phase	Spec doc
F1	WireGuard-only crypto	modern audited tunnel	helix-core WG (kernel fast path, boringtun fallback)	P0/P1	01,03
F2	QUIC obfuscation (MASQUE / RFC 9298)	WG-over-HTTP/3, looks like web	helix-transport CONNECT-UDP via quinn+h3; edge terminates :443/udp	P0/P1	01 (D1)
F3				P2	01

#	Mullvad feature	What it does	HelixVPN implementation	Phase	Spec doc
	MASQUE CONNECT-IP (RFC 9484)	native IP-over-HTTP/3 datapath	quinn+h3 CONNECT-IP path (advanced, no inner WG)		
F4	Shadowsocks obfuscation	WG-in-Shadowsocks	helix-transport Shadowsocks wrap (shadowsocks-rust core)	P2	01
F5	UDP-over-TCP	WG when UDP fully blocked	helix-transport UoT	P2	01
F6	LWO (lightweight obfs)	cheap WG signature evasion	helix-transport LWO (XOR/padding)	P1	01
F7	Automatic obfuscation	try methods until one works	client escalation ladder, driven by handshake-failure events	P1	01,03
F8	Custom WG port / 443 / 53	port-based evasion	edge multi-listener + port-hopping	P1	01
F9	DAITA (anti traffic-analysis)	constant packet size + cover traffic	optional shaping layer above WG (maybenot-style state machine)	P2	01
F10	Multi-hop	entry/exit separation	nested WG, control-plane orchestrated	P2	01,02
F11	Kill-switch	no leaks if tunnel drops	OS firewall rules driven by helix-core state machine	P1	03
F12	Split tunneling	per-app / per-route bypass	per-route AllowedIPs + per-app rules (Android/desktop)	P1	03
F13	DNS leak protection	force tunnel DNS	core sets tunnel DNS; blocks plaintext :53 off-tunnel	P1	03
F14	No-logging	no connection logs	architectural: ephemeral Redis	P1	02

#	Mullvad feature	What it does	HelixVPN implementation	Phase	Spec doc
			presence, no durable table		
F15	Anonymous account (no email)	anonymous identity	optional anonymous device-token enrollment alongside OIDC	P1	02
F16	Post-quantum handshake	PQ-safe key exchange	WG PQ pre-shared layer / ML-KEM (FIPS-203) in pki + core; hybrid never PQ-only	P2	01,02
F17	Per-device management	see/revoke devices	registry + Console; device.revoked → instant edge enforcement (<1s)	P1	02

HelixVPN-beyond-Mullvad rows (the differentiators, not present in Mullvad):

#	Feature	Implementation	Phase
X1	Multi-network overlay (1 user → N nets)	Connectors advertise CIDRs; Gateway compiles policed overlay	P1
X2	Outbound-only network onboarding	Connector reverse tunnel	P1
X3	Self-hostable Mullvad-class stack	rootless Podman pod + helixvpnctl	P1
X4	Direct P2P + NAT traversal + DERP-style relay	STUN-like discovery, hole punching, helix-relay fallback	P2
X5	8-platform reach incl. Aurora + HarmonyOS NEXT	Flutter forks + native tunnel shims	P1/P3

10. Scope per phase (in / out)

The phase spine is the CLD roadmap [04_P0], [04_P1], [04_P2], [04_ARCH §12], [SYNTHESIS §4]. Each phase's scope is a binding boundary: a feature listed **OUT** for a phase **MUST NOT** be built in that phase (it becomes a tracked workable item per §11.4.93). The phase→task→subtask decomposition lives in the later phase documents; this section fixes only the *boundary*.

10.1 Phase 0 — Spike (~3–4 weeks, throwaway bodies on production interfaces)

Goal: de-risk the make-or-break unknowns before committing to the MVP shape. Surviving artifacts are *interfaces*, not implementations: the Transport trait, the helix-wg boringtun wrapper, the orchestrator + status enum, the FFI surface [04_P0].

IN scope (P0)	OUT of scope (P0)
Exit gates G1–G6 (below)	any production-quality code; any persistence beyond a static map
Plain-UDP WG client→gw→connector LAN datapath	Console / Access / Connector <i>product</i> UIs
MASQUE/QUIC through a simulated DPI UDP block	Postgres data model, RLS, multi-tenancy
iOS NEPacketTunnelProvider Rust core under memory ceiling (G3 — make-or-break)	obfuscation transports beyond plain-UDP + one MASQUE spike
Go-vs-Rust edge benchmark (G4 decides D5)	HA, multi-region, P2P, multi-hop, DAITA, PQ
flutter_rust_bridge FFI drives core from Dart (G5)	Aurora / HarmonyOS / Windows / macOS shims
Push-based reconcile from a <i>static</i> map (G6)	event bus, policy compiler, real coordinator

Exit gates [04_P0]: **G1** plain-UDP WG client→gw→connector LAN (≥80% bare-link throughput); **G2** MASQUE/QUIC through a DPI UDP block (≥50% of plain); **G3** iOS NE Rust core under memory ceiling with ≥30% headroom (*make-or-break*); **G4** Go-vs-Rust edge benchmark → decides D5; **G5** flutter_rust_bridge FFI drives core from Dart; **G6** push-based reconcile from a static map. Test rig: Linux netns + nftables DPI sim + tc netem.

10.2 Phase 1 — MVP (self-hostable)

Goal: a genuinely self-hostable product meeting the MVP Definition-of-Done.

IN scope (P1)	OUT of scope (P1 → P2/ P3)
Go modular monolith: identity / registry / ipam / pki / policy / coordinator / events / telemetry / api / store	full transport set (Shadowsocks, UoT, hardened LWO ladder) → P2
Postgres + RLS data model (no connection/traffic tables — CI-lint enforced)	DAITA, multi-hop, direct P2P + NAT traversal, post-quantum → P2
protobuf Coordinator over Connect, server-streaming WatchNetworkMap (snapshot+deltas, peers pre-policy-filtered, need-to-know)	Windows (wireguard-nt+service) + macOS NE desktop apps → P2
coordinator (in-mem topology graph, minimal deltas, p99 < 1 s)	HA + multi-region (stateless coordinators, Patroni PG, NATS) → P2
Redis Streams event backbone	GitOps/policy-as-code → P2
identity (OIDC + anonymous device tokens) + enrollment (device-gen WG keypair, private key never leaves) + short-lived mTLS device cert + revoke < 1 s	HarmonyOS NEXT + Aurora OS builds → P3
policy compiler (Tailscale-ACL-flavored, default-deny, fail-closed → AllowedIPs + nftables/eBPF verdict maps)	WASM browser MASQUE proxy → P3
Gin REST + WS/SSE; helixvpncctl (Cobra) + Podman quadlets	billing-optional multi-tenant; third-party audit + reproducible builds → P3
Auto obfuscation ladder: plain / LWO / QUIC-MASQUE	
Access app on iOS + Android + Linux; Connector daemon; Console (web)	

MVP Definition-of-Done — 8 acceptance criteria [04_P1] (each requires captured runtime evidence per §11.4.69; none may be a metadata-only PASS):

1. self-host from zero on a clean VPS via `helixvpncctl init`;
2. enroll a Connector **and** a Client;

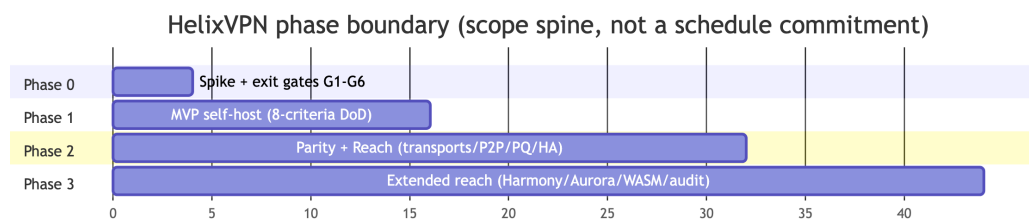
3. reach an authorized LAN host **and** deny an unauthorized one;
4. auto-escalate to MASQUE when plain WG is blocked;
5. policy edit reflected in < 1 s with no restart;
6. device revoke enforced in < 1 s;
7. kill-switch + DNS-leak protection verified (no leak on tunnel drop);
8. no durable connection log (schema-lint + runtime check) **and** all three apps drive the system.

10.3 Phase 2 — Parity + Reach

IN: full transport set (+Shadowsocks, UDP-over-TCP, hardened LWO, auto-ladder with per-network memory + regional priors); **DAITA** via maybenot; **direct P2P + NAT traversal** (STUN-like discovery, hole punching, DERP-style helix-relay fallback); **multi-hop** nested WG; **post-quantum** handshake (ML-KEM/FIPS-203 PSK, hybrid-never-PQ-only; Rosenpass as an alternative); desktop apps (Windows wireguard-nt+service, macOS NE); policy-as-code/GitOps; HA + multi-region (stateless coordinators, Patroni PG, NATS JetStream) [04_P2]. **OUT:** HarmonyOS/Aurora/WASM (→P3); managed billing (→P3).

10.4 Phase 3 — Extended reach

IN: HarmonyOS NEXT + Aurora OS builds (real native tunnel-shim work — the **biggest platform risk**), WASM browser MASQUE proxy, billing-optional multi-tenant, third-party audit + reproducible builds [04_ARCH §12], [SYNTHESIS §4]. **OUT (explicitly deferred / non-goals):** lawful-intercept, L7 inspection, ad-funded free tier — these remain product non-goals (§2.2), not "later phases".



11. Open architectural decisions D1–D8 (options + recommendation)

These are the decisions where the refined 04_VPN_CLD docs pivoted to one camp while the broader 10-LLM consensus diverged [SYNTHESIS §3]. Per §11.4.6 / §11.4.66 each is presented as **options + a recommendation**, never

silently resolved. Each Dn is **OWNER-CONFIRMABLE**; the recommendation is what the spec set proceeds on *unless overridden*, and each names the **resolution gate** (the moment evidence forces the choice).

D1 — Primary obfuscating transport

	Camp A	Camp B
Option	MASQUE/QUIC = WG-over-HTTP-3 (RFC 9298/9297/9221) — Mullvad's <i>actual</i> mechanism [04_ARCH §3.3], [04_P0], [research-masque]	Hysteria2 + Salamander (QUIC+obfs, turnkey) primary, WG fallback (plurality of the 10 analyses)
For	true Mullvad parity; single Rust impl shared client+edge (P5); "QUIC ≠ separate protocol, it IS WG-over-MASQUE" [04_ARCH §0]	ships faster (turnkey lib); battle-tested censorship circumvention
Against	more implementation surface (quinn+h3+MASQUE layer)	a <i>different</i> protocol, not WG-over-the-wire → not true parity; second crypto stack to trust

Recommendation: D1 = MASQUE/QUIC (Camp A) as the parity transport, with the escalation ladder (F7) able to *also* carry a Hysteria2-style mode as one rung. Rationale: P2 (WG core, pluggable transports) makes MASQUE the *correct* parity answer, and one Rust helix-transport crate serves client+edge (P5). Hysteria2's "ships faster" advantage is captured by adding it as a *ladder rung*, not as the architectural core.
Resolution gate: Phase-0 G2 (MASQUE through a DPI block ≥50% of plain) — if G2 fails, re-open D1 with G2 evidence.

D2 — Shared client-core language

	Camp A	Camp B
Option	Rust core + Flutter UI (plurality: 04_*,02_QWN,03_ZAI,08_CPL,10_KMI,11_MST)	Go core + Flutter UI (reuses Hysteria2's Go) (01_DSK,07_GMI)
For	wins on iOS NE memory ceiling (~15 MB) + WASM target; no GC on hot path; matches Mullvad/WARP precedent	simpler; reuses Hysteria2 Go directly; faster initial velocity
Against		

Camp A	Camp B
steeper FFI surface (flutter_rust_bridge + UniFFI)	GC + runtime risk under the iOS NE memory ceiling; weaker WASM story

Recommendation: D2 = Rust core (Camp A). The iOS NE ~15 MB ceiling is the single hardest constraint [04_ARCH §5.7]; a GC'd Go data plane is risky there, and Rust is the only choice that also reaches WASM cleanly.

Resolution gate: Phase-0 G3 (iOS NE Rust core ≥30% memory headroom — *make-or-break*). G3 is the literal decider; if Rust cannot fit, D2 *and* the whole client strategy re-open.

D3 — Event bus

	Camp A	Camp B
Option	Redis Streams (MVP) → NATS JetStream (scale) (04_P1,02_QWN,11_MST)	NATS JetStream from start (01_DSK,10_KMI)
For	honors the mandated stack [05_YB0]; one fewer moving part for self-host; bus taxonomy is transport-agnostic so the swap is a transport change	durable subjects + multi-region fan-out from day one; no later migration
Against	a later swap for fleet scale	extra infra a single-node self-hoster does not need

Recommendation: D3 = Redis Streams for MVP, NATS JetStream in Phase 2. The mandated stack [05_YB0] names Redis; the event taxonomy is bus-agnostic (events:devices, events:routes,...) so Phase-2 fleet scale is a *transport swap, not a redesign* [04_ARCH §4.3]. **Resolution gate:** Phase-2 HA/multi-region work — the bus interface is fixed in 02 so the swap is mechanical.

D4 — IP-subnet collision across N joined RFC1918 networks

	Camp A	Camp B
Option	IPv6 ULA /48 per tenant + Tailscale 4via6 mapping of advertised IPv4 LANs [04_ARCH §3.4]	CGNAT 100.64.0.0/10, 1:1 per network (07_GMI,10_KMI)
For		

	Camp A	Camp B
	clean, collision-proof; native v6 overlay; Tailscale-proven	simpler mental model; no v6 dependency on legacy clients
Against	requires v6 overlay plumbing everywhere	100.64/10 space exhaustion with many large networks; NAT bookkeeping

Recommendation: D4 = ULA /48 per tenant + 4via6 (Camp A), with per-network NAT into the connector as the documented fallback for overlapping ranges [04_ARCH §3.4]. Only 07_GMI/10_KMI/CLD actually solved this; it is a **v1 must-decide** (overlapping 192.168.1.0/24s are the default real-world case). **Resolution gate:** Phase-1 IPAM design in 02 — the overlay-addressing scheme is frozen before the registry/ipam service is built.

D5 — Gateway edge language (MASQUE termination)

	Camp A	Camp B
Option	Rust (quinn+h3, shares helix-transport byte-for-byte) [04_P0 G4]	Go (quic-go+masque-go, turnkey)
For	obfuscation logic shared byte-for-byte client(?)edge (P5); off the GC hot path	mature turnkey libs; same language as control plane
Against	one more Rust surface on the server	a <i>second</i> MASQUE impl to keep in sync with the Rust client → drift risk (violates P5)

Recommendation: D5 = Rust edge (Camp A), pending the G4 benchmark. P5 (one core, three consumers) strongly favors Rust so the wrap/unwrap code is identical on both ends. **Resolution gate:** Phase-0 G4 (Go-vs-Rust edge benchmark) is the *explicit decider* — if Go wins decisively on throughput/CPU *and* a shared-MASQUE drift mitigation is acceptable, D5 flips with G4 evidence.

D6 — Transport topology (symmetric vs asymmetric per-leg)

	Camp A	Camp B
Option	single protocol end-to-end	asymmetric per-leg: Hysteria2/QUIC user(?)gateway, WireGuard gateway(?)networks (11_MST — distinctive)
For		best-fit-per-leg: heavy obfuscation only where censorship lives (user(?)gw); plain

	Camp A	Camp B
	one transport to reason about/operate	efficient WG on the trusted gw↔connector leg
Against	obfuscation overhead on legs that don't need it	two transport regimes to operate + a translation point at the gateway

Recommendation: D6 = asymmetric per-leg as the *operating model*, expressed through the same helix-transport crate (Camp B mechanics, Camp A codebase). MST's best-fit-per-leg insight is elegant [11_MST] and falls out naturally: the gateway↔connector leg defaults to plain WG/UDP (P3 outbound, trusted), the user↔gateway leg runs the obfuscation ladder. This is *not* a second codebase — it is the same pluggable-transport crate (P2/P5) configured per-leg. **Resolution gate:** Phase-1 coordinator WatchNetworkMap design — per-leg transport policy is part of the pushed desired-state.

D7 — MVP ambition

	Camp A	Camp B
Option	lean tunnel-first (CLI between 2 hosts → API → apps) (02_QWN,01_DSK,04_P0)	full ecosystem / "Connectivity-OS" from v1 (10_KMI,03_ZAI)
For	de-risks the hard parts first; ships something usable fast; matches the Phase-0/Phase-1 split	a grander v1 story
Against	less impressive initial surface	enormous scope risk; violates §11.4.4 / iteration discipline

Recommendation: D7 = lean Phase-0 spike then the 8-criteria MVP (Camp A). This is already the §10 phase spine; the "Connectivity-OS" ambition is the *sum of the phases*, reached incrementally, not a v1 boulder. **Resolution gate:** none — this is settled as the document set's organizing principle; recorded here so the ambition is not silently re-inflated.

D8 — Licensing model

The licensing decision is **defined in full in §7.2** (options L-A/L-B/L-C + recommendation); it is registered here as D8 so the decision set is complete and greppable.

	Summary
Options	L-A AGPLv3 · L-B source-available (BSL→Apache) · L-C permissive (Apache-2.0/MIT) reusable cores + source-available managed Console (§7.2)
Recommendation	L-C — permissive reusable cores (helix-core/helix-ui/helix-proto, §11.4.28/.74) + source-available (BSL-class) for the commercial multi-tenant/managed layer
Against	a competitor could run a managed clone of the permissive core with no give-back (mitigated by the source-available managed layer)

Resolution gate: OWNER-GATED (§11.4.66) before public release; until resolved every repo carries an explicit `LICENSE.PENDING` marker, never an assumed license (§11.4.6).

11.1 Decision summary

ID	Decision	Recommendation	Resolution gate
D1	primary obfuscating transport	MASQUE/QUIC (Hysteria2 as a ladder rung)	Phase-0 G2
D2	shared client-core language	Rust core + Flutter UI	Phase-0 G3 (make-or-break)
D3	event bus	Redis Streams MVP → NATS Phase 2	Phase-2 HA work
D4	IP-subnet collision	ULA /48 + 4via6 (NAT fallback)	Phase-1 IPAM design
D5	gateway edge language	Rust (quinn+h3)	Phase-0 G4 benchmark
D6	transport topology	asymmetric per-leg via one crate	Phase-1 coordinator design
D7	MVP ambition	lean spike → 8-criteria MVP	settled
D8	licensing model (§7.2)	L-C (permissive reusable cores + source-available managed layer)	OWNER-GATED before public release

12. Glossary of normative terms

Term	Meaning in this spec set
Connector	network-side agent inside a private network; dials outbound; advertises CIDRs.
Gateway	the public VPS; control plane (Go) + data plane/edge (Rust).
Client	end-user Access app; receives an overlay IP.
Overlay IP	the stable address a node holds in the HelixVPN overlay (see D4).
Network map	the per-agent, policy-filtered desired-state pushed by the coordinator (WatchNetworkMap).
Edge	the Rust data-plane component of the Gateway terminating obfuscated transports.
Transport	a pluggable obfuscation layer <i>under</i> WireGuard (plain UDP, MASQUE, Shadowsocks, UoT, LWO).
Fail-static	if the control plane is down, existing tunnels keep forwarding (P1).
No-logging-by-construction	no durable connection/traffic/packet table exists; CI-enforced (P7).
Tenant	an isolated organization boundary (Postgres RLS); a self-hoster running networks for multiple clients has multiple tenants.
Phase boundary	the binding in/out scope per §10; a feature OUT for a phase is a tracked workable item, not built early.

13. Cross-document contracts this document fixes

The following invariants are *fixed here* and inherited (never weakened, per §11.4.35) by every later document:

1. **Three roles, two planes, fail-static** (§3, P1) → 02 control plane, edge spec.
2. **WG crypto core + pluggable transport** (P2, D1) → 01 transport spec.
3. **Rust core + Flutter UI + schema-generated clients** (P5, D2) → 03 client spec, 04 UI/proto.
4. **Push-based WatchNetworkMap, p99 < 1 s** (P4, D3, D6) → 02.
5. **No durable traffic table; control-only audit** (P7, F14) → 02 data model + CI schema-lint.

6. **8-criteria MVP DoD + phase boundaries** (§10) → phase docs; each becomes §11.4.93 workable items.
7. **Mullvad-parity matrix as the acceptance checklist** (§9) → every feature doc traces a row.
8. **Open decisions D1–D8 carry recommendations + resolution gates** (§11; D8 = licensing, defined in §7.2) → each gate is a Phase-0/1/2 task or OWNER-GATED; re-opening requires new evidence (§11.4.6/.7).

13.1 Helix-ecosystem submodule wiring (must be designed in, not retrofitted)

The source research predates the now-incorporated submodules/ [SYNTHESIS §8]. This document records the binding so later docs cannot omit it:

Submodule	Role in HelixVPN	Constitution anchor
containers (vasic-digital)	the mandated container orchestration layer for deploy + on-demand integration-test infra	§11.4.76, §11.4.161
helix_qa + challenges	anti-bluff QA / Challenge layer for the 8-criteria DoD + parity matrix	§11.4.27, §11.4.5/.69/.107
docs_chain	mechanizes spec-doc + workable-items sync (this doc set)	§11.4.106
security	security tooling (P7, §7 hardening)	§7
vision_engine	video-evidence QA for UI/ connection-state flows	§11.4.107/.158
llm_*/panoptic	mark <i>not-applicable</i> unless a concrete need arises (no speculative wiring)	§11.4.6

Sources verified

- 04_VPN_CLD/HelixVPN-Architecture-Refined.md §0–§14 ([04_ARCH]) — primary product/architecture source.
- 04_VPN_CLD/HelixVPN-Phase0-Spike.md ([04_P0]), HelixVPN-Phase1-MVP.md ([04_P1]), HelixVPN-Phase2-Parity.md ([04_P2]), HelixVPN-helix-ui-Flutter.md ([04_UI]).

- 05_VPN_YB0.md ([05_YB0]) — operator mandate brief (stack + platform + two-way/multi-network requirements).
- 00_VPN_Initial_Res.md, 01_VPN_DSK.md, 02_VPN_QWN.md, 03_VPN_ZAI.md, 06_VPN_GRK.md, 07_VPN_GMI.md, 08_VPN_CPL.md, 09_VPN_GCT.md, 10_VPN_KMI.md, 11_VPN_MST.md — the 10 independent analyses cross-referenced for D1–D7 camps.
- v09-research/_SYNTHESIS.md ([SYNTHESIS]) — cross-document synthesis + decision register.
- [research-masque] — RFC 9298 (CONNECT-UDP), RFC 9297 (HTTP Datagrams), RFC 9221 (unreliable QUIC datagrams), RFC 9484 (CONNECT-IP); Mullvad QUIC-mode = WG-over-MASQUE.

Constitution bindings applied: §11.4.44 (revision header), §11.4.6 (no-guessing — open decisions carry options+recommendation, never silent picks), §11.4.66 (decision discipline), §11.4.17/.35 (universal vs project, inheritance), §11.4.76/.161 (containers/rootless), §11.4.93 (phases→workable items), §11.4.106 (docs_chain sync), §11.4.65/.153 (HTML+PDF[+DOCX] exports follow in refinement).